Monetary Policy and Personal Income Distribution: A Survey of the Empirical Literature

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Motivation

The topic of growing inequality has gained attention from policymakers due to its adverse economic effects.

Academic research suggests that inequality is linked to hindered growth and increased risk of financial crises. (Ostry, et al. 2014; Bordo and Meissner 2012; Rajan 2010)

Another critical topic is the continuous use of unconventional MPs –involves purchases of assets mainly being held by the rich - during the COVID-19 outbreak

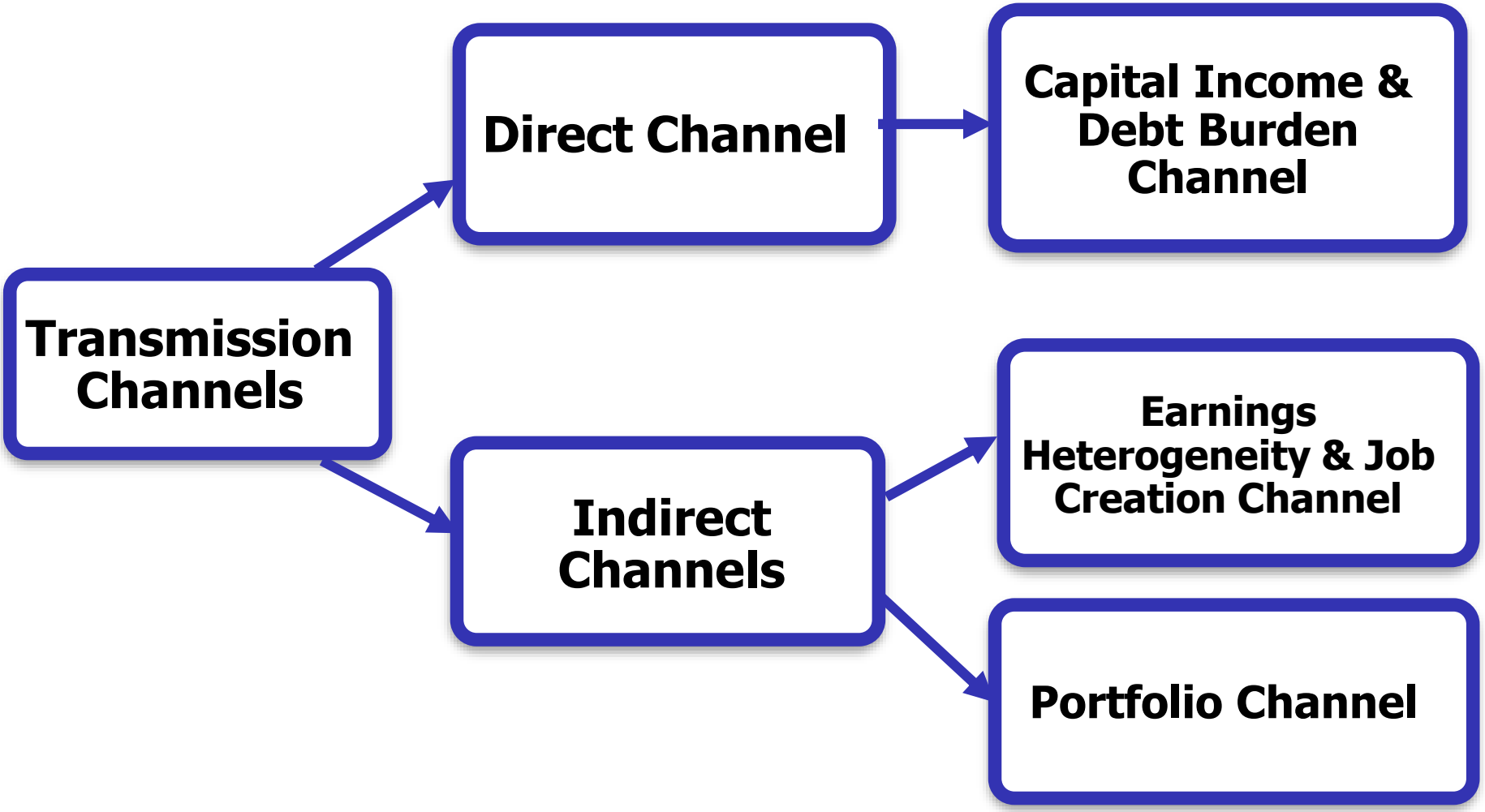
The continuous use of UMPs raised concerns about distributional effects of these policies ,resulting in a growing body literature (IMF, Central Banks)

The **objective** of this paper is to conduct a survey of the recent literature that evaluates, empirically, the distributional impacts of monetary policy.

There are many channels through which monetary policy can affect income distribution

Start by the basic definition of **Net Financial Income:**

$$\begin{aligned} \textit{Net Financial Income} = \textit{Wages} + \textit{Interest receipts} \\ + \textit{Business profits} + \textit{Transfers} \\ + \textit{Capital Gains} - \textit{Debt payments} \end{aligned}$$


Transmission Channels
Direct Channel
Capital Income & Debt Burden Channel
Indirect Channels
Earnings Heterogeneity & Job Creation Channel
Portfolio Channel

Capital Income and Debt Burden Channel

A decrease in the interest rate => a decrease in capital income and debt payments

The overall impact depends on **individual's net wealth**

- A lower-class household with a mortgage-financed house will benefit (lower payments & no financial wealth upon which interest is received)
- A rich individual is probably worse-off since she/he is likely to hold bonds and has little debt.
- The overall impact of income distribution of an expansionary policy is to reduce income inequality.

Earnings Heterogeneity & Job Creation Channel

An expansionary monetary policy increases output and reduces unemployment.

The overall effect depends on the **response of low-skilled and high-skilled workers to the change in the unemployment rate**.

In countries where high-skilled workers grow relatively more than low-skilled countries, an expansionary policy increases the inequality

In countries where low-skilled workers are the first to be unemployed in a slump and the first to be hired on the recovery, an expansionary policy reduces the inequality.

Portfolio Channel

A monetary policy changes asset prices, leading to capital gains/losses

The overall effect depends on the **distribution of assets among income percentiles** and on the **degree by which each asset class responds to changes in interest rates.**

If equity prices rise more than housing prices after a given interest rate cut, inequality increases.

If equities are more unevenly distributed amongst households, an expansionary monetary policy would increase inequality.

Empirical Strategies: Key Points

1. decide how to measure the monetary policy change and income inequality

2. choose what type of monetary policy change to be considered

3. decide on the empirical strategy and econometric techniques to be used

Measures of the variables

1. Changes in MP

Conventional MP

- short-term interest rates
- shadow interest rates (case of periods with zero lower bound)

Unconventional MP

- FED Quantitative Easing of the ECB's asset purchase program
- monetary base
- size of the monetary authority's balance sheet.

2. Inequality

Gini coefficient

- to measure gross income (pre-tax, pre-transfer income) or net income (post-tax, post-transfer income)

90/10 or 80/20 ratio

- the ratio between the incomes of the top decile and the lowest decile

Type of Monetary Policy Change

Surprise or Unexpected Shock

Sign restrictions

(Hafemann, et al.,2018; Mumtaz and Theophilopoulou, 2017 ; Uhlig, 2005)

External instruments

(Herradi and Leroy, 2019; Stock and Watson, 2018)

Residual of the regression of the federal funds rate, GDP deflator and unemployment

(Coibion et al.,2017; Davtyan, 2016)

Forecast errors for interest rate, GDP and inflation

(Furceri, et al. .2018; Auerbach and Gorodnichenko, 2013)

Two broad empirical strategies

Three-step approach (3SLS)

- deals with unconventional MP

First step: VAR model to estimate the aggregate effects of monetary policy on economic variables.

Second step: Logit/Probit models to distribute aggregate changes into individual-level data

Third step: Individual changes are re-aggregated in inequality measures to be compared

Using readily disposable data

- deals with conventional MP

-data on Gini coefficients or percentiles ratios

-uses VAR models (Bayesian VAR, Panel VAR, Structural VAR) or Local Projections

Literature Review on MP and income inequality

1. Studies dealing with unconventional MP
2. Studies dealing with conventional MP
3. Studies dealing with both conventional and unconventional MP

1. Studies dealing with the distributional effects of Unconventional MP

Saiki and Frost (2014)	Montecino and Epstein (2015)	Casiraghi et al. (2018)	Lenza and Slacalek (2018)	Bunn et al. (2018)
• VAR model • Japan 2008-2014 • Effect of central bank assets to nominal GDP on 80/20 ratio • An increase in the assets increases inequality	• RIF model • US 2008-2014 • Effect of QE on 90/10 ratio • An increase in QE increases inequality	• 3-step approach • Italy 2010 • Effect of ECB unconventional MP on Gini coefficient and 90/10 ratio • An increase in ECB reduces inequality negligibly	• 3-step approach • France, Germany, Italy and Spain 1999- 2016; US 2008-2013 • Impacts of ECB UMP on Gini coefficient • An increase in ECB reduces inequality	• 3-step approach • UK 2008-2014 • Impacts of UMP on Gini coefficient • UMP has negligibly reduced income and wealth inequality

2. Studies dealing with the distributional effects of conventional MP

Furceri et al. (2018)	Hafemann et al. (2018)	Aye et al. (2019)	Davtyan (2016)	Villarreal (2014)
• Local Projections • 32 emerging countries 1990-2013 • Effect of short-term interest rate on GINI coefficient • Unanticipated contractionary monetary policy increases income inequality • Asymmetric effects of unanticipated MP	• VAR model • 6 countries 1990-2015 • Effect of shadow interest rates on Gini coefficient • Interest rate reduction increases inequality. • Degree of redistribution after taxes and transfers is important.	• Local projections • US 1980-2008 • Effect of federal funds rate on Gini coefficient and 90/10 ratio • Contractionary monetary policy shock increases income inequality • MP impact is higher under low uncertainty.	• VAR model • US 1979-2012 • Effect of federal funds rate on Gini coefficient • Contractionary monetary policy shock decreases income inequality	• Local Projections • Mexico 1995-2012 • Effect of overnight interbank lending rate on GINI coefficient • Contractionary monetary policy shock reduces income inequality

3. Studies dealing with the distributional effects of both MPs

Guerello (2018)

- VAR model
- 17 Euro Area countries 2001-2015
- Effect of short-term interest rate and ECB assets on Theil coefficient of income
- Expansionary conventional shocks reduce income inequality
- The effect of unconventional MP depends on the share of deposits on total financial assets
- The higher the share, the more the reduction in inequality.

Inui et al. (2017)

- Local Projections
- Japan 1981-1998
- Effect of short-term interest rates on log of Gini coefficient and 90/10 ratio for labor earnings, pre-tax income, disposable income, consumption, expenditure and financial positions
- Expansionary shocks have negligible effect on the inequality measures

Hesary et al. (2019)

- VECM model
- Japan 2002-2017
- Effect of short-term interest rates/ MB on 90/10 ratio
- Contractionary shocks increase income inequality

Discussion

Mixed results

- The evidence tends more towards that expansionary monetary policy decreases inequality, or that contractionary monetary policy increases income inequality.

Conventional Monetary Policy

- Case of expansionary policies, three panel data studies covering 43 countries suggest that such measures generally decrease income inequality.
- Case of contractionary policies, with a focus on some works, mostly single-country studies, the evidence indicates an increase in income inequality for most countries

Unconventional Monetary Policy

- This policy is exclusively expansionary
- The evidence is less conclusive. The impact of UMP appears to be contingent on the specific country context; A reduction in inequality in Europe but an increase in the US and Japan.

Discussion (cont'd)

Roles of Fiscal Policy Redistribution & Labor Share

- Shaping the ultimate impact of monetary policy on inequality
- Higher levels of redistribution diminish the impact of monetary policy on inequality
- A higher labor share intensifies the impact of monetary policy on inequality.

Asymmetric Effects of Monetary Policy

- Increases in interest rates have statistically significant effects on income distribution, whereas the effects of interest rate reductions are not statistically different from zero.
- This contradicts the conventional view that distributional effects of interest rate changes are temporary and tend to balance out over the business cycle.

Concluding Remarks

For conventional monetary policy, the strongest evidence suggests that expansionary MP decreases inequality, while contractionary MP increases it.

For unconventional MP, methodologies indicate that UMP generally reduces inequality, although there is some contradictory evidence, case of Japan and US.

There is consistent evidence that higher fiscal policy redistribution is associated with lower inequality impacts of monetary policy and higher wage share is associated with higher impacts

Recommendations for Future Research

Advocate for a more distributive fiscal policy and emphasizes the need for a precise estimation of the asymmetry between interest rate hikes and falls.

Study the interactions between fiscal and monetary policies on income distribution and other dimensions of distributive impact, as gender and race

Consider methodological robustness and samples' length to avoid the misconceptual belief that the distributional effects of monetary policy diminish over time, a belief which contradicted with much empirical evidence.